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5-hydroxymethyl-2-furaldehyde impairs *Candida albicans* - *Staphylococcus epidermidis* interaction in co-culture by suppressing crucial supportive virulence traits

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ABSTRACT

Polymicrobial biofilms involving fungal-bacterial interactions are stated to modulate host immune response and exhibit enhanced antibiotic resistance. In this milieu, clinically important opportunistic pathogens *Candida albicans* and *Staphylococcus epidermidis* associate synergistically and instigate implant and blood stream infections. Impediment of virulence traits that support successive pathogenic lifestyle and inter-kingdom interactions without altering the microbial growth represents an attractive alternate strategy. To accomplish this objective, 5-hydroxymethyl-2-furaldehyde (5HM2F), a reported antibiofilm agent against *C. albicans*, was considered for this study. 5HM2F significantly repressed the biofilm formation of *S. epidermidis* and mixed-species at 300 μg/mL and 400 μg/mL, respectively without modulating the growth. Microscopic analyses and phenotypic assays explicated the competency of 5HM2F to impede biofilm formation, hyphal growth, initial attachment, intercellular adhesion, and fungal-bacterial interaction. Further, 5HM2F greatly reduced the secreted hydrolases production. Reduced content of biofilm matrix components upon 5HM2F treatment was believed to be the underlying reason for enhanced antibiotic and/antifungal susceptibility. Additionally, qPCR analysis correlated well with *in vitro* bioassays wherein, 5HM2F was identified to repress the expression of important genes associated with hyphal morphogenesis, adhesion, biofilm formation and virulence in both mono-species and mixed-species. Reduced virulence and colonization of mono-species and mixed-species in 5HM2F treated *Caenorhabditis elegans* substantiated the antibiofilm and antivirulence potential of 5HM2F. Overall, this study proposes 5HM2F as a potent therapeutic candidate against single and mixed-species biofilm infections of *C. albicans* and *S. epidermidis*.

1. Introduction

Antibiotics linger as the most preferred choice of treatment for many common infections owing to the ease of accessibility. However, the inappropriate and/or overuse of antibiotics has driven the unfavorable consequences of resistance development and incompetence of antibiotic treatments in clinical settings over past few decades [1–5]. Besides, formation of microbial biofilm on biotic and abiotic surfaces is stated to intensify the degree of resistance development [6,7]. Biofilm is cocooning of microbes in self-generated extracellular polymeric substances (EPS) that is poised with nucleic acids, proteins, carbohydrates, and lipids [8]. Microbial planktonic-biofilm switching is a multifactorial process, which is reported to be accompanied by the modification of gene expression and virulence production [9–11]. Consequently, microbes adopt the tendency to resist the influence of host defense system and antibiotic treatments via attainment of transitory multicellular lifestyle [9,10,12,13].

Irrespective of the clinical relevance of polymicrobial biofilms, studies pertaining to mixed-species biofilms are less common. Polymicrobial biofilms hold importance as they could act as infectious reservoir for diverse microbes [14]. In this regard, *Candida albicans*, an important opportunistic fungal pathogen of blood stream infections is stated to cause polymicrobial infections in association with many Gram positive and Gram-negative bacteria. Particularly, in life-threatening blood stream infections, it has been identified to be frequently co-isolated with *Staphylococcus* spp. among which, coagulase negative

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Staphylococcus epidermidis ranks first [15]. Further, infections instigated by their partnership have been experimentally recorded to considerably amplify the death rates of nematodes than mono-species infections [16]. Also, this fungal-bacterial interaction is stated to increase morbidity and mortality rates of underweight neonates with late onset sepsis [17].

In mixed state, *S. epidermidis* augments the resilience of biofilm by exhibiting affinity to yeast and hyphal forms of *C. albicans*. Particularly, the association of *Staphylococcus* spp. with invasive hyphal form of *C. albicans* is reported to aid invasive lifestyle of *Staphylococcus* and complicate the severity of infections [18,19]. Moreover, Adam et al. [6] have reported the reduced susceptibility of mixed-species biofilms to both antifungals and antibiotics. Precisely, the symbiotic fungal-bacterial interaction during infections is stated to append benefits of increased virulence and enhanced resistance towards antibiotics and host innate immunity [6,18,20–22]. Hence, there is a crucial need to probe alternate strategies for confronting notorious inter-kingdom interaction of *C. albicans* and *S. epidermidis*.

Lately, antivirulence therapy has been demarcated to act as one of the alternate strategies, which reduces the feasibilities of resistance development by particularly mitigating the pathogenicity of microbes without influencing their metabolic rates [23]. In an earlier research from our group, we have reported that 5-hydroxymethyl-2-furaldehyde (5HM2F) originating from marine bacterium; *Bacillus subtiltis* tumbles the virulence of *C. albicans* [24]. Additionally, this natural furan compound has also been recounted to execute many beneficial biological effects [25–29]. With this background, this study was carried out as an extension of our previous work [24] to study the antibiofilm and antivirulence activity of 5HM2F against single and mixed-species of *C. albicans* and *S. epidermidis* using *in vitro* bioassays and simple eukaryotic model, *Caenorhabditis elegans*.

2. Materials and methods

2.1. Microbial strains and culture conditions

In the current study, fungal reference strain *C. albicans* ATCC 90028 and bacterial reference strain *S. epidermidis* ATCC 35984 were used. *C. albicans* was maintained in yeast extract peptone dextrose (YEPD) (1% yeast extract, 2% peptone and 2% dextrose) agar (1.8%) plates and cultured regularly in YEPD broth. For biofilm assays, spider broth (1% mannitol, 0.2% dipotassium hydrogen phosphate and 1% nutrient broth) was used for facilitating hyphal development of *C. albicans* [24]. On the other hand, *S. epidermidis* was maintained on tryptone soy broth (TSB) agar (1.8%) plates and regularly cultured in TSB. The broth cultures were incubated in shaker incubator (160 rpm) at 37 °C. For all liquid culture assays, an assay period of 24 h and 48 h was maintained for *S. epidermidis* and *C. albicans*, respectively. Overnight cultures of *C. albicans* and *S. epidermidis* were diluted to 0.8 OD and 0.4 OD, respectively for adjusting the cell density to $\sim 1X10^8$ CFU/mL. The adjusted cultures were used as standard cell suspensions of mono-species for performing all experiments. Mixed-species culture was prepared by mixing equal parts of standardized cell suspensions of each microorganism just prior to the experiments. TSB was used as the optimal medium to support growth and biofilm formation of mixed-species [6] with optimal incubation period of 48 h for all assays.

2.2. Impact of 5HM2F on biofilm formation of S. epidermidis and mixed-species

A 40 mg/mL stock of 5HM2F (Sigma Aldrich, Switzerland) was prepared in methanol and refrigerated till further use at 4 °C. The influence of 5HM2F on biofilm formation was studied using crystal violet (CV) staining assay in 24 well microtitre plate (MTP) [24]. One milliliter of TSB comprising different concentrations of 5HM2F (100–500 μg/mL) and 1% of standard cell suspensions of *S. epidermidis* and mixed-species were added to each well of MTP separately. Plain TSB and TSB with methanol (vehicle) were considered as blank and control, respectively. Further, the plates were incubated in static condition at 37 °C. Post incubation, free floating and slackly adhered cells from each well were removed and biofilms were washed with sterile water for three times. One milliliter of 0.4% CV was added to each well for staining the biofilms. Surplus and unbound CV stain was removed from wells after 5 min, which was followed by brief washing of wells with sterile water for three times. Then, biofilm adhered CV was eluted by adding 1 mL of 10% glacial acetic acid to the dried wells. The least concentration of 5HM2F that impedes more than 80% of biofilm formation was taken as minimum biofilm inhibitory concentration (MBIC). The optical density (OD) of eluted CV stain was measured at 570 nm to estimate the biofilm formation of test pathogens in the absence and presence of 5HM2F. The percentage of inhibition was calculated using the following formula:

$$\% \text{ of biofilm inhibition} = \{([\text{Control OD}_{570\text{nm}} - \text{Treated OD}_{570\text{nm}}]/\text{Control OD}_{570\text{nm}})*100\}$$

2.3. Influence of 5HM2F on growth and metabolic viability of S. epidermidis and mixed-species

The impact of 5HM2F on growth and cell viability was estimated using broth microdilution and alamar blue (AB) assay, respectively [30]. In brief, wells comprising 1 mL of TSB were added with various concentrations of 5HM2F (100–500 μg/mL) and 1% standard cell suspensions of *S. epidermidis* and mixed-species. Suitable blanks and 5HM2F free controls were also maintained. After incubation, OD of control and 5HM2F treated cultures was recorded at 600 nm. Subsequently, cells were collected using centrifugation and suspended in sterile phosphate buffered saline (PBS) after brief washing. Then, AB assay was performed by adding 0.35 μM of AB dye (Sigma Aldrich, Switzerland) to each sample. After 4 h of incubation in dark condition, the fluorescence intensity of oxidized (blue and non-fluorescent)/reduced (pink and fluorescent) form of AB dye was quantified in a multimode reader using excitation and emission wavelength of 560 and 590 nm, respectively.

2.4. Microscopic studies

Biofilm architecture of control and 5HM2F treated samples was further analyzed using microscopic techniques such as light microscope (LM), confocal laser scanning microscope (CLSM) and scanning electron microscope (SEM) [31]. Initially, *S. epidermidis* and mixed-species biofilms were allowed to form on glass slides (1 cm^2), which were kept in MTP wells holding 1 mL of suitable media, 1% inoculum and 5HM2F (300 μg/mL for *S. epidermidis* and 400 μg/mL for mixed-species). After incubation, biofilms bound to the glass slides were washed with sterile water to remove slackly adhered cells and glass slides were subsequently processed according to the technique employed. For LM observation, biofilms were stained with 0.4% CV, washed after 5 min using sterile water and air-dried. Then, control and 5HM2F treated biofilms were visualized under LM (Nikon Eclipse 80i, USA) at 400X magnification. For CLSM analysis, biofilms were stained with 0.1% acridine orange in dark condition, de-stained after 5 min and visualized under CLSM (LSM 710, Carl Zeiss, Germany). For in-depth visualization of biofilm architecture, SEM (VEGA 3 TESCAN, Czech Republic) analysis was done by initially fixing the biofilms formed on the glass slides using 2.5% glutarladehyde for 1 h. Then, samples were dehydrated for 2 min in various concentrations of ethanol (20, 40, 60, 80 and 100%), which was followed by air-drying and gold sputtering prior to analysis.

2.5. Effect of 5HM2F on aggregation and adherence of mono-species and mixed-species

2.5.1. Auto-aggregation and co-aggregation

To estimate the aggregation rate, cells were initially harvested by centrifugation (10000 rpm for 10 min) from mono-species cultures grown in the absence and presence of 5HM2F (at MBIC). The collected cells were then suspended evenly in 5 mL of sterile PBS. In case of co-aggregation, equal parts of mono-species cultures were mixed well for a total volume of 5 mL. Before subjecting the cell suspensions to standing incubation, 2 mL of culture was collected from each tube and their absorbance was measured at 600 nm. Then, tubes were kept undisturbed till the control cells aggregate completely at the bottom of the tube. After incubation, 2 mL of cell suspensions were collected from top layer of each tube without any disturbance and absorbance was again recorded at 600 nm [30,32].

2.5.2. Cell adherence

Impact of 5HM2F on cell adherence was assessed using the protocol mentioned by Arrecubieta et al. [33] with modifications. Control and 5HM2F (at MBIC) treated cells of mono-species and mixed-species were collected by centrifugation (10000 rpm for 10 min) and suspended aseptically in sterile PBS. One milliliter of each sample was then separately added to the wells of MTP. After 2 h of incubation, unbound cells were discarded and wells were subsequently washed twice using sterile water and air-dried. Then, adherence rate was estimated by CV staining method as mentioned earlier.

2.5.3. Hyphal-bacterial attachment assay

To assess the effect of 5HM2F on hyphal-bacterial attachment, a method reported by Peters et al. [20] was followed with modifications. *C. albicans* was allowed to form biofilm in the wells of MTP holding 1 mL of hyphal inducing spider broth. After incubation for 48 h at static condition, unbound and loosely adhered cells were discarded followed by thorough washing of wells with sterile water for three times. Then, the control and 5HM2F (at MBIC) treated *S. epidermidis* was adjusted to 0.4 OD at 600 nm. One milliliter of OD adjusted culture was added to the wells with pre-formed hyphal biofilms of *C. albicans*. Plate was incubated at 37 °C with gentle shaking (120 rpm) for 1 h. Post incubation, non-adherent cells were removed and cells adhered to the wells were washed twice with sterile water. Subsequently, 1 mL of sterile PBS was added to each well and well bound cells were collected by vortex-mixing the wells briefly for a minute followed by scraping of remaining adherent cells using cell scraper. After that, collected cells were serially diluted and spread plated on mannitol salt agar plates. Number of colonies formed on Staphylococcal differential media was manually calculated after 24 h of incubation to estimate the total CFU/mL of control and 5HM2F treated *S. epidermidis* adhered to *C. albicans* hyphal biofilm.

2.6. Impact of 5HM2F on other virulence traits of mono-species and mixed-species

2.6.1. Yeast to hyphal transition

To examine the impact of 5HM2F on morphological transition of *C. albicans* in mixed state, spider agar plates comprising 1% fetal bovine serum (FBS) were prepared in the absence and presence of 5HM2F (at MBIC). Subsequently, plates were spot inoculated at the center using 2 μL standard cell suspension of mixed-species and incubated at 37 °C [24]. After 48 h of incubation, plates were observed for filamentous protrusions around the colony and documented using gel documentation system (GelDoc XR+, Bio-Rad).

2.6.2. Protease production

Protease production of *S. epidermidis* and mixed-species was assessed using two different substrates i.e. casein and bovine serum albumin (BSA). Firstly, TSB agar plates supplemented with 1% casein in the absence and presence of 5HM2F (at MBIC) were added with 2 μL of standard cell suspensions at the center of the plate to assess the caseinase activity [34]. In parallel, agar plates supplemented with 0.05% $MgSO_4$, 0.1% KH_2PO_4, 1% BSA and 2% glucose in the absence and presence of 5HM2F (at MBIC) were added with 2 μL of standard cell suspensions at the center of the plate to estimate the level of proteolysis of BSA [35]. The plates were observed for white opaque zone around the colony after 48 h of incubation at 37 °C and subsequently documented.

2.6.3. Phospholipase production

For qualitative estimation of phospholipase production of *C. albicans* in mixed state, 8% of egg yolk emulsion was used as substrate. Briefly, YEPD agar plates with and without 5HM2F (at MBIC) were added with substrate, 0.5 mM $CaCl_2$ and 1 M NaCl. Then, 2 μL standard cell suspension of mixed-species was used to spot inoculate at the center of control and treated plates [24]. The plates were observed for precipitation zone around the colony after 48 h of incubation at 37 °C and subsequently documented.

2.6.4. Lipase production

To quantify the lipolytic activity of mono-species and mixed-species, substrate mixture was initially prepared by mixing solution I (0.3% p-nitrophenyl palmitate dissolved in 2-propanol) and solution II (0.1% gummi arabicum and 0.2% sodium deoxycholate dissolved in 50 mM Na_2Po_4 buffer (pH- 8)) at 1:9 ratio. Then, the supernatant of control and 5HM2F (at MBIC) treated mono-species and mixed-species cultures grown up to appropriate optimal incubation period were collected by centrifugation (10000 rpm for 10 min). Nine parts of substrate mixture was added to one part of supernatant and consequently incubated for 1 h in dark condition. Post incubation, reaction mixtures were centrifuged (12000 rpm for 5 min) and added immediately with 1 mL of 1 M Na_2CO_3 to halt further reaction. The absorbance was recorded at 410 nm [36] and percentage of lipase inhibition was calculated using the following formula.

$$\% \text{ of lipase inhibition} = \{([\text{Control OD}_{410nm} - \text{Treated OD}_{410nm}]/\text{Control OD}_{410nm})*100\}$$

2.7. Estimation of matrix components of control and 5HM2F treated mono-species and mixed-species biofilms

To quantify biofilm components, mono-species and mixed-species biofilms were allowed to form in 6 well MTP in the absence and presence of 5HM2F (at MBIC). After incubation, non-adherent cells were removed and washed twice with sterile water. The content of each well was then scraped off completely and collected separately in fresh tubes using 200 μL of Tris-EDTA buffer (10 mM each of Tris and EDTA, pH-8). Standard phenol-sulfuric acid method and phospho-vanillin method were employed for quantifying total carbohydrate and lipid contents, respectively [30,37,38]. For estimating extracellular DNA (eDNA) and protein contents, control and treated samples were initially vortex-mixed for 5 min and the supernatants comprising extricated biofilm matrix were collected by centrifugation (12000 rpm for 5 min). Standard Bradford method was employed for quantifying biofilm matrix associated protein content. For assessing eDNA content, agarose gel electrophoresis (AGE) was performed in 1.2% agarose gel using equal quantity of control and 5HM2F treated supernatants and consequently documented using gel documentation system (GelDoc XR+, Bio-Rad).

2.8. Antibiotic and/antifungal susceptibility testing

To appraise the antibiotic and/antifungal susceptibility of *S. epidermidis* and mixed-species in the presence of 5HM2F, antibiotics such as gentamycin, rifampicin, vancomycin and tetracycline (Hi-Media Laboratories, India) and an antifungal, ketoconazole (SRL, India) were used. Minimum inhibitory concentration (MIC) of test antibiotics and antifungal was initially determined against *S. epidermidis* and *C. albicans* using broth microdilution assay [39,40]. MIC is defined as the least concentration of drug that impedes visible growth after 24 h of incubation. Agar well diffusion method was employed for assessing the impact of 5HM2F on antibiotic and/antifungal susceptibility. Standard cell suspensions of *S. epidermidis* and mixed-species were evenly swabbed on TSB agar plates with and without 5HM2F (at MBIC) using sterile cotton swabs. At the center of the each plate, well of ~5 mm diameter was punched. In case of *S. epidermidis* swabbed plates, MIC of each test antibiotic was loaded separately in the wells of control and 5HM2F treated plates. In case of mixed-species swabbed plates, two combinations i.e. ketoconazole + rifampicin (K + R) and ketoconazole + tetracycline (K + T) were used. Precisely, equal ratio of test antifungal and antibiotic at 0.5X MIC, 1X MIC and 2X MIC were added to the wells of three discrete sets of control and 5HM2F treated plates. Additionally, 5HM2F (at MBIC) was added to the wells of *S. epidermidis* and mixed-species swabbed plates to evaluate the antimicrobial effect of individual 5HM2F. HiAntibiotic zone scale (Hi-Media Laboratories, India) was used to determine the zone diameter of antimicrobial activity in control and 5HM2F treated plates following 24 h of incubation at 37 °C.

2.9. Gene expression studies using real-time PCR (qPCR)

Total RNA content was extracted from mid log phase cultures of mono-species and mixed-species grown in the absence and presence of 5HM2F (at MBIC) by following the method described by Oh and So [41]. The sequestered RNA transcripts were further reverse transcribed to complementary DNA (cDNA) using High Capacity cDNA Reverse Transcription Kit (Applied Biosystems, USA). The details of biofilm and virulence specific primers of *C. albicans* (*ras1*, *cph1*, *efg1*, *als3*, *eap1*, *hwp1*, *ume6* and *tup1*) and *S. epidermidis* (*agrA*, *icaA*, *icaD*, *sdrG*, *sdrF*, *sdrH*, *bhp*, *ebh*, *aap* and *atlE*) are provided in Table 1. Then, cDNA samples were added to gene specific primers that were mixed separately with SYBR Green master mix (Applied Biosystems, USA) at a predefined ratio. Two house-keeping genes were included as internal controls i.e. *its* (internal transcribed spacer) gene for *C. albicans* and *rplU* (50S ribosomal protein) for *S. epidermidis.* The optimal number of cycles and thermal profile was titrated according to the manufacturer's guidelines. Gene expression pattern of control and 5HM2F treated samples was analyzed using real-time PCR (7500 sequence detection system, Applied Biosystems, USA). After appropriate normalization of cycle threshold (Ct) values of control and treated samples with respective endogenous controls (ΔCt), comparative threshold method ($2^{-\Delta\Delta Ct}$) was utilized for appraising the gene expression pattern of mono-species and mixed-species in the absence and presence of 5HM2F [52].

2.10. In vivo validation of antivirulence activity of 5HM2F (at MBIC) using Caenorhabditis elegans model

The simple eukaryotic model, *C. elegans* was employed for affirming the antivirulence efficacy of 5HM2F (at MBIC). Maintenance of nematodes was done in accord with the standard methodologies of Brenner [53]. Nematodes were fed with overnight culture of *Escherichia coli* OP50 (prescribed laboratory food source of *C. elegans*) that is pre-adjusted to ~$1X10^6$ CFU/mL. L4 stage nematodes (n = 10) taken in 1 mL of M9 buffer (0.1% 1 M $MgSO_4$, 0.3% KH_2PO_4, 0.5% NaCl and 0.6% Na_2HPO_4) were infected separately with *S. epidermidis* and mixed-species in the absence and presence of 5HM2F (at MBIC). Regular assessment of nematodes' survivability was done up to 96 h to study the antivirulence potential of 5HM2F.

To assess *in vivo* colonization, six groups of worms (~20 numbers per group) were infected separately with *C. albicans, S. epidermidis* and mixed-species in the absence and presence of 5HM2F (at MBIC). Nematodes fed with laboratory food source were included as control group. After 12 h of infection, non-adherent microbes were removed by two washes. Then, microbial burden in infected and 5HM2F treated groups was assessed using CFU assay. Worms were crushed using mini-homogenizer to extract colonized microbes. Samples were serially diluted and plated on mannitol salt agar (specific medium for *Staphylococcus*) and candida differential agar (specific medium for *Candida*) to enumerate the number of bacterial and fungal cells in infected and 5HM2F treated groups [46].

2.11. Statistical analysis

All experiments were carried out in three biological and experimental replicates. Data were depicted as mean ± standard deviation. Student's T test and one way ANOVA were performed using SPSS software package (Chicago, IL, USA) to determine the statistical significance between control and 5HM2F treated samples.

3. Results

3.1. Non-microbicidal antibiofilm activity of 5HM2F against S. epidermidis and mixed-species

CV staining method unveiled the ability of 5HM2F to mitigate biofilm formation of *S. epidermidis* and mixed-species in a dose dependent

Table 1
List of *C. albicans* and *S. epidermidis* specific primers used in this study.

S. No.	Organism	Primer	Forward primer (5′-3′)	Reverse primer (5′-3′)	Length	Reference
1.	*C. albicans*	*ras1*	CCCAACTATTGAAGATTCTTATCGTAAA	TCTCATGGCCAGATATTCTTCTTG	106	[42,43]
2.		*efg1*	GCCTCGAGCACTTCCACTGT	TTTTTTCATCTTCCCACATGGTAGT	85	[44,45]
3.		*cph1*	TATGACGCTTCTGGGTTTC	ATCCCATGGCAATTTGTTGT	163	[44,46]
4.		*hwp1*	GCTCCTGCTCCTGAAATGAC	CTGGAGCAATTGGTGAGGTT	184	[43,47]
5.		*eap1*	TGTGATGGCGGTTCTTGTTC	GGTAGTGACGGTGATGATAGTGACA	66	[45,48]
6.		*als3*	CAACTTGGGTTATTGAAACAAAAACA	AGAAACAGAAACCCAAGAACAACC	80	[45,49]
7.		*ume6*	GAACAATGGTGGTGGTAGTGG	CCTACAACCTAAACAGCTTAA	179	[47]
8.		*tup1*	CTTGGAGTTGGCCCATAGAA	TGGTGCCACAATCTGTTGTT	175	[43,50]
9.	*S. epidermidis*	*icaA*	TTGATGACGATGCGCCTTTT	CTGCAAGAGATTGACTTCGCT	171	[30]
10.		*icaD*	GACAGAGGCAATATCCAACGG	ACAAACAAACTCATCCATCCGA	223	[30]
11.		*bhp*	TGATGACAACGCAACGACAA	TGGTGTTGGACTCGTAGCTT	178	[51]
12.		*aap*	GGGCAAACGTAGACAAGGTC	GCTTTCGCTTCATGGCTACT	153	[51]
13.		*sdrG*	GTGACTTGCCTCCTGAAAAA	TCCGGTGTTTCGAATGTAAT	232	[51]
14.		*sdrF*	TGAAAAAGAGAAGACAAGAACCA	GATTGTCTTCAGCCGCTTTA	168	[51]
15.		*sdrH*	AAAAAGCCATTTTTGTTCCA	CATACGAATCAACCCCAAAG	200	[30]
16.		*ebh*	CTAAAGGAACATGGGCAGGC	AAACACCCCAGTTGCTAGGA	195	[30,51]
17.		*atlE*	ATAGAAACGGTGTGGGACGT	ACCTGCACCCCAAGATAAGT	185	[51]
18.		*agrA*	TGTAACCAGTCACAGTGAGCT	CCCCGCTTTAACTCAATCGT	186	[30]

manner (Fig. 1). 5HM2F at 300 μg/mL and 400 μg/mL hampered a maximum of 88% and 85% of biofilm formation of *S. epidermidis* and mixed-species, respectively. Unaltered growth and metabolic viability of control and 5HM2F treated *S. epidermidis* and mixed-species as witnessed in broth microdilution and AB assays (Fig. 2) exemplified the non-microbicidal nature of 5HM2F at all tested concentrations. With these interpretations, 300 μg/mL and 400 μg/mL of 5HM2F was fixed as MBIC against *S. epidermidis* and mixed-species, respectively and taken forward for performing successive experimentations.

3.2. Microscopic revelation of antibiofilm effect of 5HM2F

The biofilm architecture of 5HM2F treated slides was found to be slack and abridged than that of the control slides (Fig. 3). The light microscopic images clearly revealed substantial reduction of biofilm formation of *S. epidermidis* and mixed-species upon treatment with 300 μg/mL and 400 μg/mL of 5HM2F, respectively. CLSM ortho images revealed a great reduction in the thickness and surface coverage area of biofilms in treated slides than that of the control slides. SEM images of *S. epidermidis* mono-species biofilm affirm that the biofilm cells were less aggregated and monolayered in 5HM2F treated slide, whereas in control slide, biofilm cells were highly aggregated and multilayered. High adherence of aggregated *S. epidermidis* cells to *C. albicans* was clearly witnessed in control slide. However, the interaction was found to be greatly disturbed upon treatment, which is evident from the gap (indicated by red arrows) observed between *C. albicans* and *S. epidermidis* cells in SEM image of 5HM2F treated mixed-species biofilm.

3.3. 5HM2F mitigates initial adherence, aggregation, and hyphal-bacterial attachment

Overall, 5HM2F (at 300 μg/mL for *S. epidermidis* and at 400 μg/mL for *C. albicans* and mixed-species) treatment considerably reduced initial adherence, aggregation and hyphal-bacterial attachment. Initial adherence of 5HM2F treated *C. albicans*, *S. epidermidis* and mixed-species to MTP wells was considerably reduced up to 48.7%, 52% and 41.6%, respectively than untreated control (Fig. 4A). Additionally, 5HM2F treatment was found to impede 83%, 78% and 71% aggregating ability of *C. albicans*, *S. epidermidis* and mixed-species, respectively than respective controls (Fig. 4B). Interestingly, adherence of 5HM2F treated *S. epidermidis* to *C. albicans* hyphal biofilm was mitigated up to 69% than that of untreated *S. epidermidis*, which suggests the potential of 5HM2F to impede *C. albicans* - *S. epidermidis* interaction (Fig. 4C).

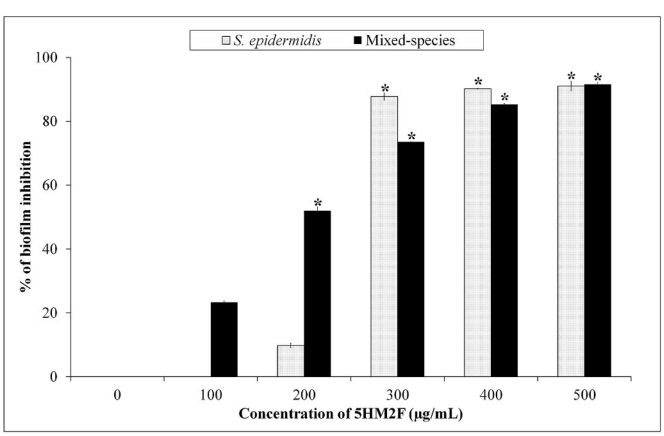

Fig. 1. Bar graph representing the impact of 5HM2F (100–500 μg/mL) on biofilm formation of *S. epidermidis* and mixed-species. Error bars indicate standard deviation. * indicates statistical significance between control and 5HM2F treated samples ($p < 0.005$).

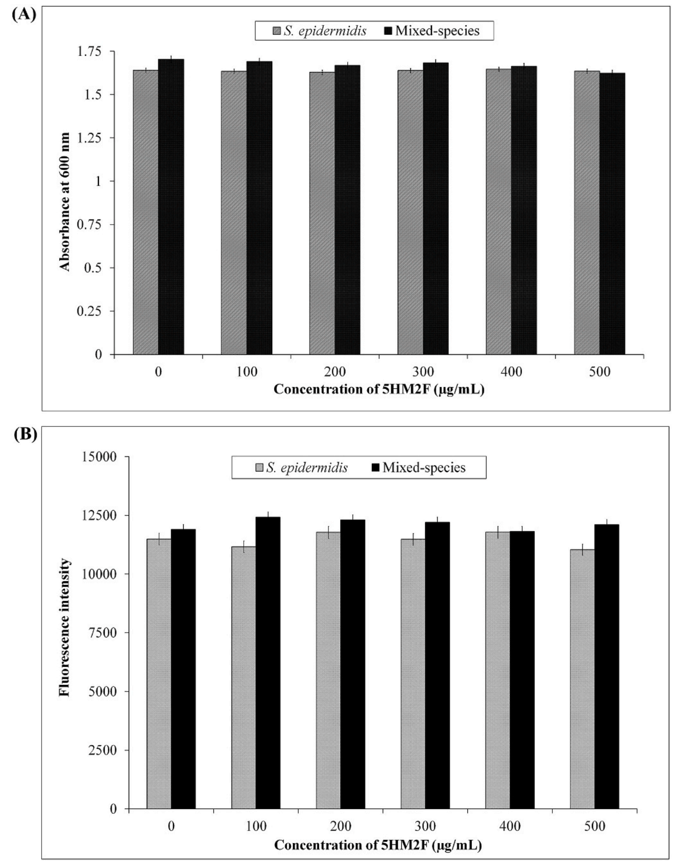

Fig. 2. Bar graphs depicting the influence of 5HM2F (100–500 μg/mL) on (A) growth and (B) metabolic viability of *S. epidermidis* and mixed-species. Error bars indicate standard deviation.

3.4. 5HM2F impairs morphological transition and secreted hydrolases production

The filamentation of mixed-species in 5HM2F (at 400 μg/mL) treated plate was found to be highly mitigated than control plate (Fig. 5A). Further, 5HM2F (at 300 μg/mL for *S. epidermidis* and at 400 μg/mL for *C. albicans* and mixed-species) treatment considerably decreased the production of protease, phospholipase and lipase. The zone diameter of caesinase activity in 5HM2F treated *S. epidermidis* (18 mm) and mixed-species (18.5 mm) plates was found to be reduced than that of *S. epidermidis* (30.5 mm) and mixed-species (30 mm) control plates. On the other hand, *S. epidermidis* was deficient in cleaving BSA, which could be perceived from the absence of proteolysis zone around the bacterial colony in control and 5HM2F treated plates. However, the BSA cleaving ability of *C. albicans* was found to be undisturbed in mixed-species, which could be observed from the white opaque zone developed around the control colony. The zone diameter of proteolysis of BSA in 5HM2F treated mixed-species (5 mm) plate was greatly reduced than that of control (14 mm) plate (Fig. 5B). The zone diameter of phospholipase activity of mixed-species in 5HM2F treated (19 mm) plate was less than that of control (26 mm) plate (Fig. 5C). Additionally, lipolytic activity of *C. albicans*, *S. epidermidis* and mixed-species was reduced up to 71%, 78% and 45%, respectively in the presence of 5HM2F (Fig. 5D).

3.5. 5HM2F thwarts matrix components of mono-species and mixed-species biofilms

Carbohydrate content of *C. albicans*, *S. epidermidis* and mixed-species biofilms was decreased up to 37%, 54% and 63%, respectively upon 5HM2F treatment. On the other hand, 5HM2F treatment has resulted in

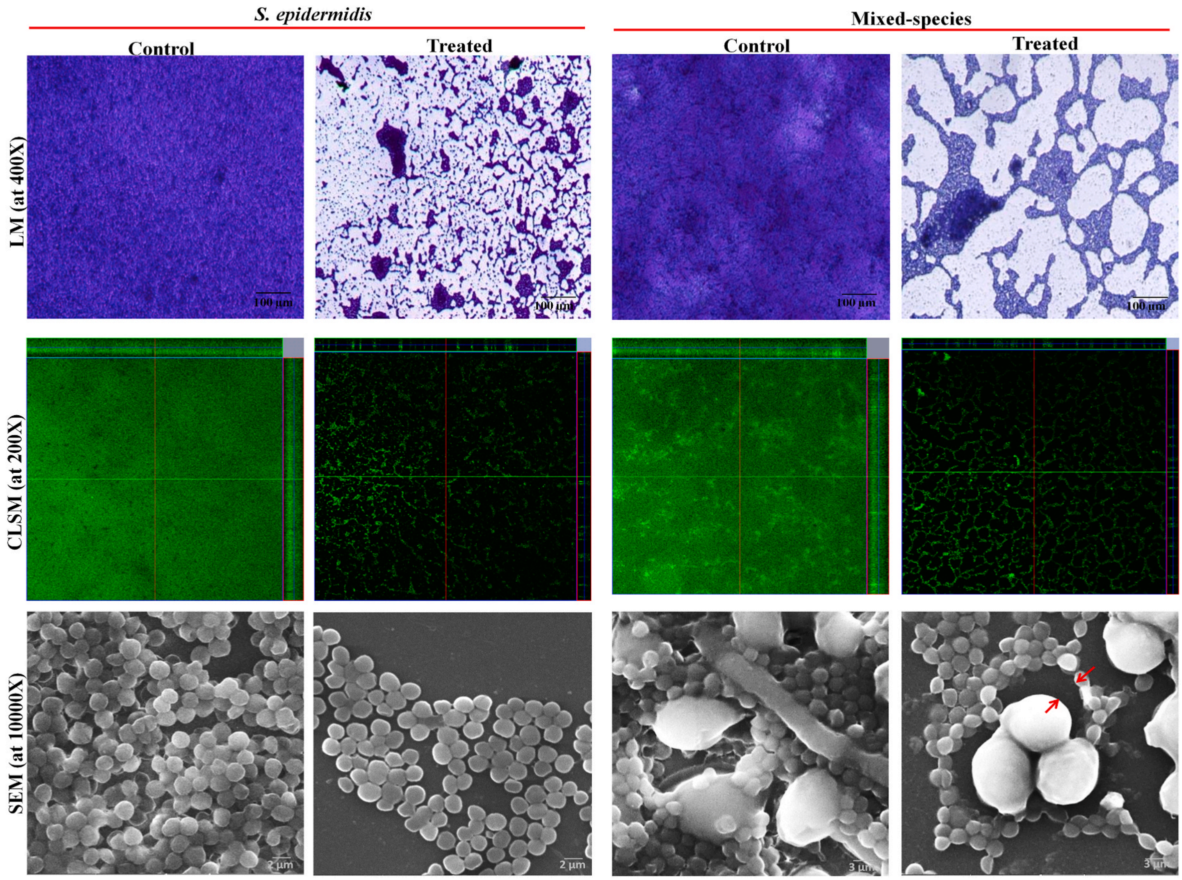

Fig. 3. Microscopic observation of biofilms. LM, CLSM and SEM images depicting the reduced biofilm formation of *S. epidermidis* and mixed-species upon treatment with 300 μg/mL and 400 μg/mL of 5HM2F, respectively. Red arrows in SEM image of 5HM2F treated slide indicate impaired *C. albicans* – *S. epidermidis* interaction upon 5HM2F treatment. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

the reduction of 26.31%, 24.17%, 38.75% protein content and 68.26%, 42.45% and 61.65% lipid content of *C. albicans*, *S. epidermidis* and mixed-species biofilms, respectively (Fig. 6A). Further, AGE divulged the ability of 5HM2F to considerably reduce eDNA content of mono-species and mixed-species biofilm matrices than that of respective untreated controls (Fig. 6B).

3.6. 5HM2F enhances the antibiotic and/antifungal susceptibility

For all test antibiotics, the zone diameter of antibacterial activity in *S. epidermidis* swabbed plates was found to be increased in the presence of 5HM2F (Fig. 7A). In mixed-species, K + R combination was observed to yield clear zones of antimicrobial activity in control plates. On the other hand, K + T combination was found to yield hazy zones in control plates, which indicates the incomplete antimicrobial activity against mixed-species. However, in 5HM2F treated plates, K + T combination yielded clear zones, which indicates the ability of 5HM2F to potentiate the antimicrobial activity of K + T combination. In both combinations, the zone diameter of antimicrobial activity was found to be increased in 5HM2F treated plates than that of control plates (Fig. 7B and C). Additionally, no zone of antimicrobial activity was observed around the wells loaded with 5HM2F alone, which implies the non-lethal effect of 5HM2F against *S. epidermidis* and mixed-spices (Fig. 7D and E). The zone diameter of antimicrobial activity in the absence and presence of 5HM2F is provided in Table 2.

3.7. 5HM2F represses the expression of hyphal morphogenesis, biofilm and virulence associated genes

Modulation in expression level of biofilm, hyphae and virulence associated genes in mono-species and mixed-species upon 5HM2F treatment was studied using qPCR analysis. For *C. albicans*, transcriptional factors that regulate hyphal growth and biofilm formation (*cph1*, *ume6*, *efg1*), hyphal specific genes (*hwp1*, *ras1*), genes involved in adhesion and fungal-bacterial interaction (*eap1*, *als3*) and a transcriptional repressor of filamentation (*tup1*) were considered. The expression level of candidate genes such as *ras1*, *cph1*, *efg1*, *ume6*, *hwp1*, *als3* and *eap1* were found to be down regulated upon 5HM2F treatment in *C. albicans* and mixed-species, thereby corroborating the antihyphal, antibiofilm and antivirulence potential of 5HM2F. Further, 5HM2F treatment was found to up regulate *tup1* (Fig. 8A), which further affirms the antihyphal activity of 5HM2F. For *S. epidermidis*, genes involved in virulence factor production (*agrA*), polysaccharide intercellular adhesion (PIA) synthesis (*icaA* and *icaD*), biofilm accumulation (*aap* and *bhp*) and adhesion to abiotic and biotic surfaces (*atlE*, *ebh*, *sdrH*, *sdrG*, and *sdrF*) were considered. Herein, 5HM2F treatment was found to considerably reduce the expression level of all candidate genes in both *S. epidermidis* and mixed-species (Fig. 8B), which attests the anti-adherence, antiaggregation, antibiofilm and antivirulence potential of 5HM2F witnessed in *in vitro* bioassays.

3.8. In vivo revelation of antibiofilm and antivirulence potential of 5HM2F

The antibiofilm and antivirulence potential of 5HM2F was further

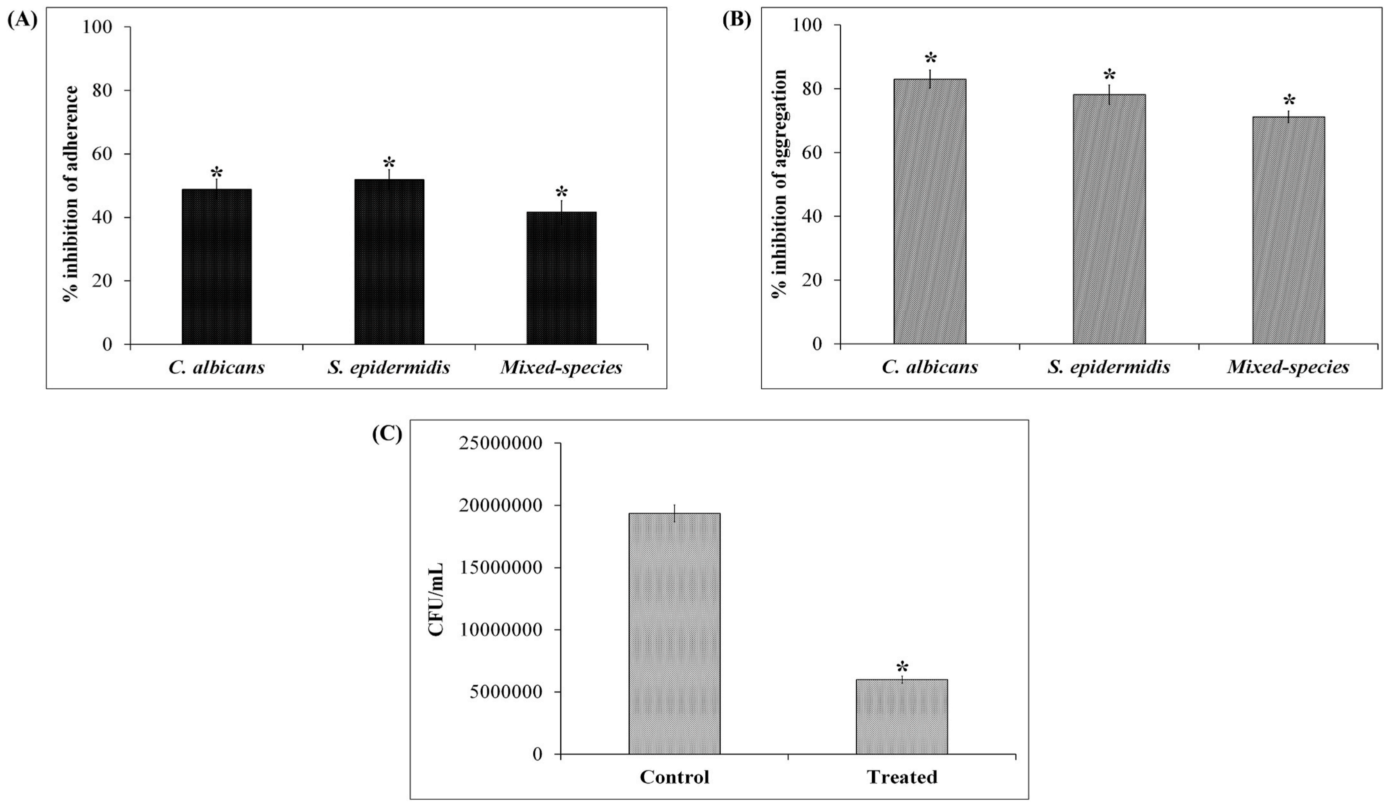

Fig. 4. Effect of 5HM2F (at 300 μg/mL for *S. epidermidis* and at 400 μg/mL for *C. albicans* and mixed-species) on (A) microbial adherence to MTP wells, (B) auto-aggregation and co-aggregation and (C) hyphal-bacterial attachment. Error bars indicate standard deviation. * indicates statistical significance between control and 5HM2F treated samples ($p < 0.005$).

assessed using simple eukaryotic nematode model, *C. elegans*. Worms infected with *S. epidermidis* and mixed-species were observed to survive only up to 6% and 3%, respectively. Interestingly, the endurance level of *S. epidermidis* and mixed-species infected worms was witnessed to be improved up to 71% and 76%, respectively in the presence of 5HM2F (Fig. 9A), which attests the antivirulence potential of 5HM2F. In furtherance, the impact of 5HM2F on *in vivo* colonization of *C. albicans*, *S. epidermidis* and mixed-species was assessed using CFU assay in which, the microbial burden in 5HM2F treated group was found to be substantially reduced than infected groups (Fig. 9B). These observations explicate the antibiofilm and antivirulence potential of 5HM2F *in vivo*.

4. Discussion

According to National Institute of Health, 65–80% of human infections are of biofilm origin, which is chronicled to exhibit increased tolerance to antibiotics & host defense system and worsen clinical outcomes [14]. Despite the treatment challenges posed by mono-microbial biofilms, enhanced drug tolerance of polymicrobial biofilms [6] is in the peril of posing additional challenges in the management of infectious diseases. These facts accentuate the necessity of probing alternative strategies. In this regard, the present study adopted antivirulence therapy as an alternative strategy to explore potent antagonistic agent against biofilm and virulence of notorious human opportunistic pathogens, *C. albicans* and *S. epidermidis* in single and mixed-state. Besides, this study was performed as an extension of the previous report [24] from our group, wherein, 5-hydroxymethyl-2-furaldehyde (5HM2F) of marine bacterial origin has been reported to exhibit proficient antibiofilm and antivirulence activity against *C. albicans*. Further, 5HM2F is naturally present in honey, edible bamboo mushroom and heat treated foods, which has been previously investigated for its anti-tyrosinase [25], anti-sickle cell anemia [26,27], anti-cancer [54] and anti-quorum sensing [55] properties. Nonetheless, to the best of the authors' knowledge, 5HM2F has not been previously scrutinized for its antagonist effect against mixed-species biofilm formed by *C. albicans* and *S. epidermidis*.

With this background, the current study initially examined the effect of 5HM2F on biofilm. In parallel to the report by Subramenium et al. [24], a dose dependent antibiofilm effect of 5HM2F was witnessed against *S. epidermidis* and mixed-species (Fig. 1). In furtherance, the influence of 5HM2F on microbial growth was assessed, as non-microbicidal effect is described as a typical feature of an ideal antivirulence agent. Since non-lethal drugs are anticipated to exclude the generation of Darwinian selection pressure on microbes, the feasibilities of microbes to resist these drugs are stated to be meager [56,57]. Accordingly, non-microbicidal effect of 5HM2F as confirmed by broth microdilution and alamar blue assays (Fig. 2) underscores the meager/nil chances of resistance development against 5HM2F. From biofilm and growth assays, 300 μg/mL and 400 μg/mL of 5HM2F was fixed as MBIC against *S. epidermidis* and mixed-species respectively, which resulted in >80% biofilm inhibition in a non-microbicidal manner. For *C. albicans*, 400 μg/mL was previously reported to be the MBIC of 5HM2F [24] and the same MBIC was adopted in this study also.

LM, CLSM and SEM analyses authenticated the antibiofilm efficacy of 5HM2F wherein, microbial adherence to glass surface, cell aggregation, biofilm biomass and biofilm surface coverage area were found to be greatly decreased in 5HM2F treated slides than the respective control slides (Fig. 3). Interestingly, enlarged view of control and 5HM2F treated mixed-species biofilm in SEM explicated the potential of 5HM2F to profoundly impair the synergistic interaction of *C. albicans* and *S. epidermidis*. Initial adherence of microbial cells to a substratum is stated to be a crucial and first step of biofilm formation in bacteria as well as fungi [58,59]. Additionally, cell to cell adhesion aided by aggregation is recounted to mediate biofilm development and

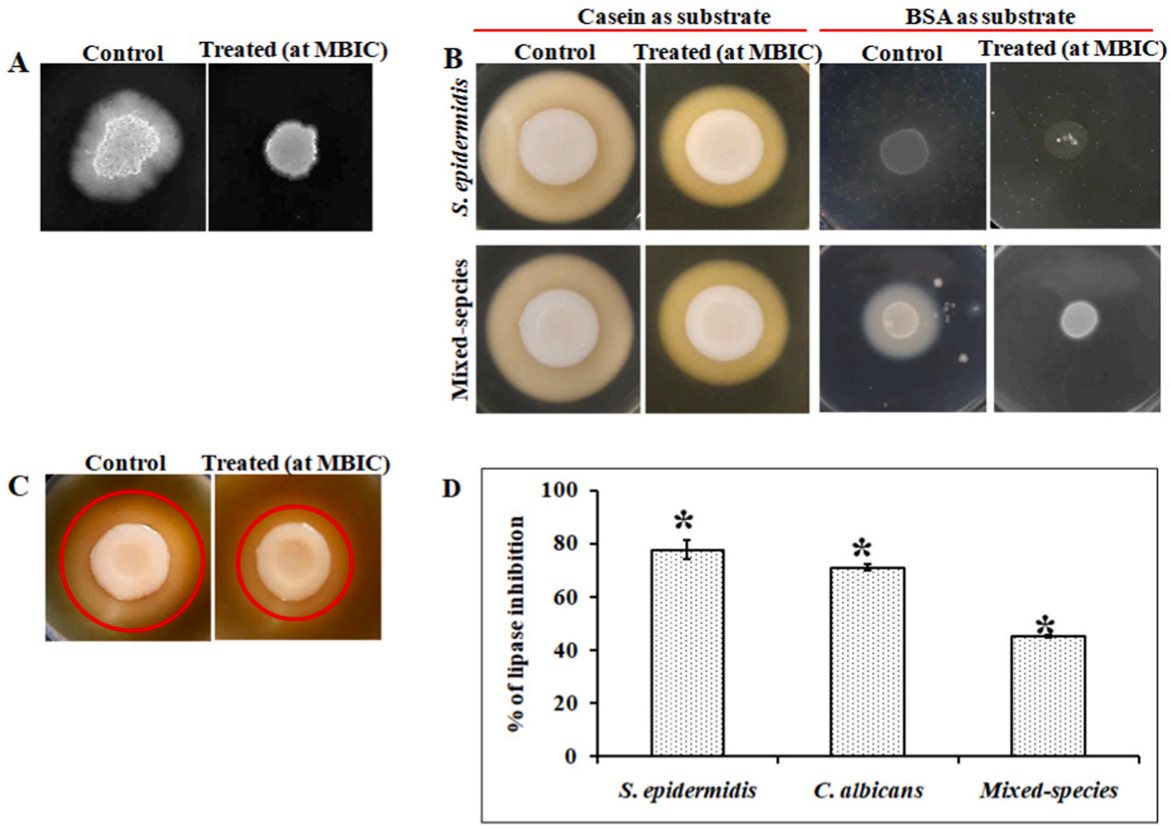

Fig. 5. Impact of 5HM2F (at 300 μg/mL for *S. epidermidis* and at 400 μg/mL for *C. albicans* and mixed-species) on (A) yeast to hyphal transition of mixed-species, (B) protease production of *S. epidermidis* and mixed-species in casein and BSA agar plates, (C) phospholipase production of mixed-species (red circle denotes the zone of phospholipase activity) and (D) lipolytic activity of mono-species and mixed-species. Error bars indicate standard deviation. * indicates statistical significance between control and 5HM2F treated samples ($p < 0.005$). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

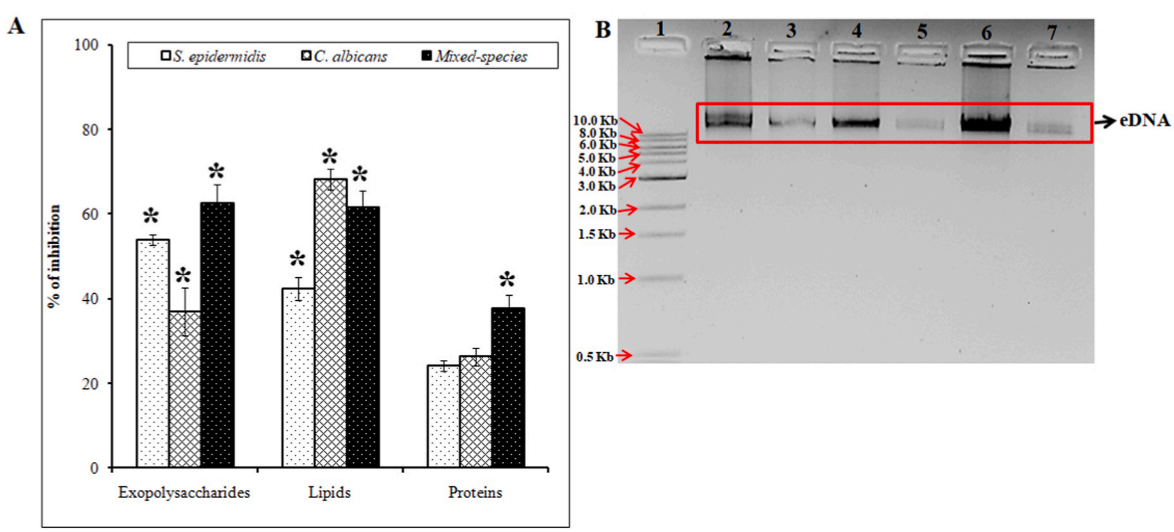

Fig. 6. Effect of 5HM2F on biofilm matrix components of mono-species and mixed-species biofilms. (A) Bar graph representing the inhibitory effect of 5HM2F on carbohydrate, protein and lipid contents of mono-species and mixed-species biofilms. Error bars indicate standard deviation. * indicates statistical significance between control and 5HM2F treated samples ($p < 0.005$). (B) Agarose gel image showing the effect of 5HM2F on eDNA content of mono-species and mixed-species biofilm. Lane 1- 1 Kb ladder; Lane 2- *S. epidermidis* Control; Lane 3- *S. epidermidis* treated with 5HM2F (at MBIC); Lane 4- *C. albicans* Control; Lane 5- *C. albicans* treated with 5HM2F (at MBIC); Lane 6- Mixed-species Control; Lane 7- Mixed-species treated with 5HM2F (at MBIC).

mixed-species interaction [60,61]. Accordingly, CV staining assay and microscopic studies endorsed the ability of 5HM2F to inhibit biofilm formation by greatly affecting initial microbial adherence, cell to cell adhesion and aggregation. To further attest the antiadherence and antiaggregation properties of 5HM2F, cell adherence, autoaggregation and co-aggregation assays were performed. Herein, reduced microbial

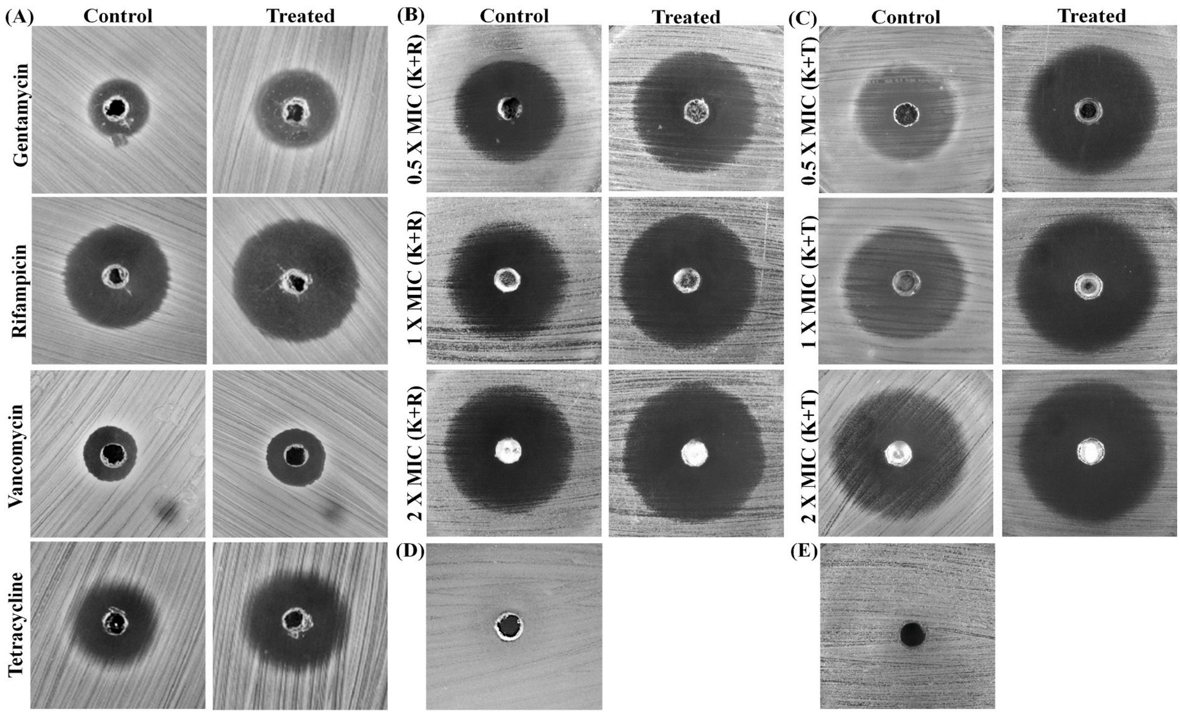

Fig. 7. Effect of 5HM2F on susceptibility of (A) *S. epidermidis* to antibiotics and mixed-species to (B) ketoconazole + rifampicin (K + R) and (C) ketoconazole + tetracycline (K + T) combinations. The impact of 5HM2F alone on (D) *S. epidermidis* and (E) mixed-species growth.

Table 2
Zone diameter of antibiotic and/antifungal activity in control and 5HM2F treated *S. epidermidis* and mixed-species.

S. No.	Organism	Antibiotic and/antifungal	Concentration (μg/mL)	Zone diameter in control plate (mm)	Zone diameter in 5HM2F treated plate (mm)
1.	*S. epidermidis*	Gentamycin	120	12.5	16
2.		Rifampicin	0.5	22	25
3.		Vancomycin	10	12	15
4.		Tetracycline	1	18	22
5.	Mixed-species	Ketoconazole + Rifampicin (K + R)	75 + 0.25	30	39
6.			150 + 0.5	33	42
7.			300 + 1	36	43.5
8.		Ketoconazole + Tetracycline (K + T)	75 + 0.5	27	34
9.			150 + 1	30	36.5
10.			300 + 2	32	38

adherence and aggregation rates upon 5HM2F treatment (Fig. 4A and B) provide additional evidences for the antagonistic potential of 5HM2F against initial attachment, aggregation, biofilm formation and interaction of *C. albicans* and *S. epidermidis*.

Amid initial attachment and aggregation, morphological transition of *C. albicans* from yeast to hyphal form is considered as an important feature in biofilm progression [59]. Besides, hyphal form is construed as a highly virulent form, which promotes invasive mode of *C. albicans* and exacerbates the infectious state [62]. Anti-hyphal activity of 5HM2F against *C. albicans* mono-species was previously testified by Subramenium et al. [24]. Thus, yeast to hyphal transition assay was performed against mixed-species. Degenerated hyphal growth upon 5HM2F treatment (Fig. 5A) suggests the ability of 5HM2F to retain its anti-hyphal activity against *C. albicans* in mixed state. Importantly, *S. epidermidis* is reported to demonstrate high adhesive force with hyphal form of *C. albicans* than yeast form [63], which highlights the importance of *C. albicans* hyphae in mixed-species interaction. Accordingly, the antihyphal potential of 5HM2F is speculated to be the key underlying factor for impaired *C. albicans* – *S. epidermidis* interaction observed in SEM micrographs of 5HM2F treated mixed-species biofilm. In addition, association of *S. epidermidis* with hyphal form of *C. albicans* is believed to support host invasion of *S. epidermidis* [18]. Decreased attachment of 5HM2F treated *S. epidermidis* to hyphal biofilm as confirmed by hyphal-bacterial attachment assay (Fig. 4C) indicates the ability of 5HM2F to thwart the invasiveness of *S. epidermidis*. Furthermore, this observation also indirectly implies the efficacy of 5HM2F to interfere with the adhesive molecules of *S. epidermidis* that supports interaction with *C. albicans*.

Besides biofilm formation, virulence arsenals of *C. albicans* and *S. epidermidis* include secretion of hydrolases such as proteases, lipases, and phospholipases, which facilitate their successful pathogenic and invasive lifestyle by aiding host matrix degrading ability. Several independent investigations have also divulged that the inactivation of genes associated with secreted hydrolases production is positively correlated with reduced virulence *in vivo* [64–66]. In view of that, reduced secreted hydrolases production upon 5HM2F treatment (Fig. 5B, C, 5D) indicates the potential of 5HM2F to mitigate the virulence level of *S. epidermidis* and *C. albicans* in both single and mixed state.

EPS matrix is another indispensable part of microbial biofilm, which defines the fate of matrix-encased sessile cells by regulating its water content, hydrophobicity, density, charge, porosity, and mechanical stability. It is also stated to confer resistance by minimizing the protrusion of host defense and drug molecules through its intricate layer. Further, this scaffolding biofilm matrix is recounted to be a composite of biopolymers such as carbohydrates, nucleic acids, proteins, and lipids [67,68]. Hence, reduced amount of biopolymers in 5HM2F treated mono-species and mixed-species biofilms (Fig. 6) represents the incompetence of cells to generate adequate slimy layer in the presence of 5HM2F for scaffolding sessile cells and forming well-defined biofilm architecture. Consequently, formation of impaired biofilm upon 5HM2F treatment is speculated to expose and sensitize the microbes to host defense system and antibiotics.

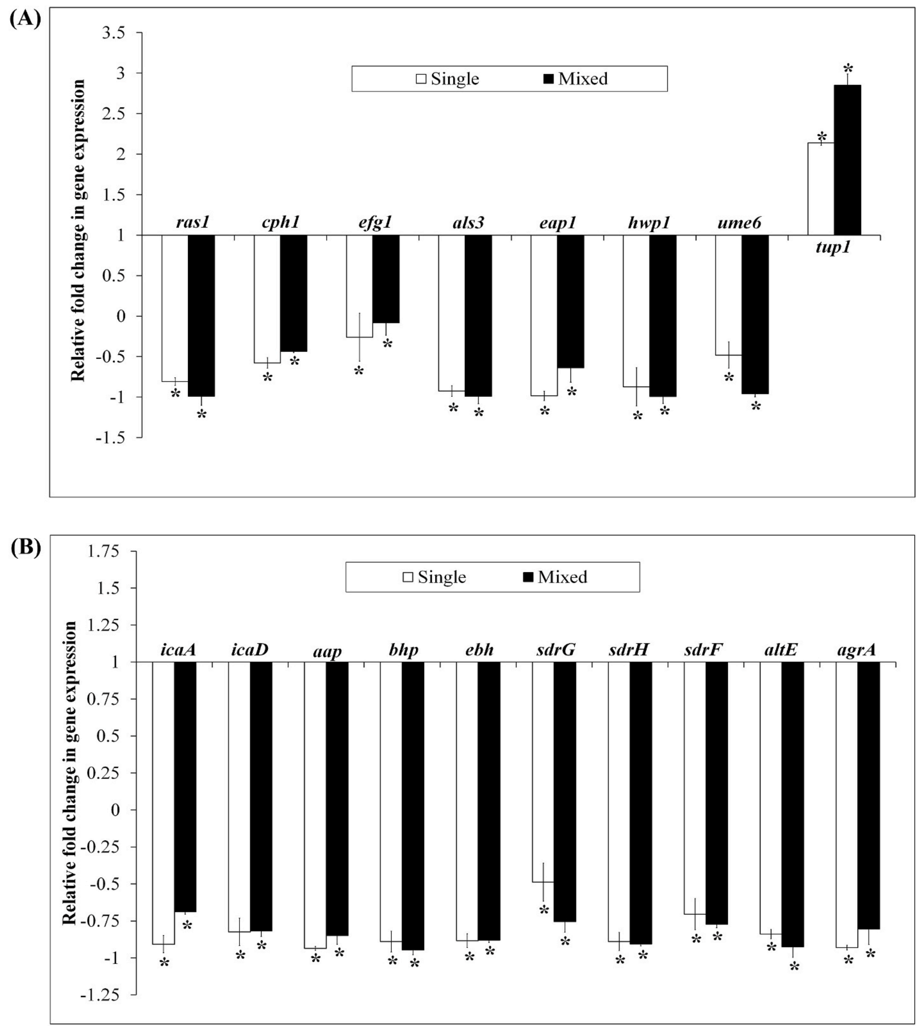

Fig. 8. Effect of 5HM2F on gene expression pattern of (A) *C. albicans* specific genes and (B) *S. epidermidis* specific genes in both single and mixed-state. Error bars indicate standard deviation. * indicates statistical significance between control and 5HM2F treated samples ($p < 0.005$).

Concomitantly, the effect of 5HM2F on antibiotic and/antifungal susceptibility was assessed further. In *S. epidermidis*, the presence of 5HM2F was found to improve the antibacterial activity of test antibiotics as predicted. Further, 5HM2F was previously identified to embellish the activity of azole antifungals such as ketoconazole, clotrimazole and miconazole against *C. albicans*. In mixed-species, the antimicrobial activity of K + R and K + T combinations was found to be considerably enhanced in the presence of 5HM2F. Noticeably, 5HM2F was proficient in potentiating the activity of K + T combination, which demonstrated incomplete antimicrobial activity (hazy zones) in the absence of 5HM2F (Fig. 7). Taken together, these results propose the feasibility of employing 5HM2F along with antibiotics/antifungals for managing mixed-species infections in clinical settings. Further, the enhanced antimicrobial activity of conventional antibiotics/antifungals in combination with 5HM2F is speculated to eradicate established biofilms of single and mixed-species.

To unravel the plausible mode of antibiofilm action of 5HM2F, qPCR analysis was performed. Hyphal morphogenesis, biofilm and virulence associated genes were considered in case of *C. albicans* and mixed-species. In *C. albicans*, *ras1* is stated to be an important regulator of virulence determinants, hyphal morphogenesis, and metabolic adaptation. The significance of *ras1* signaling in *C. albicans* pathogenesis was previously studied by Leberer et al. [69] wherein, *ras1* mutant was witnessed to be deficient in demonstrating high virulence *in vivo*. Further, *ras1* is reported to transduce environmental cues favoring yeast to hyphal transition through cAMP protein kinase A and mitogen-activated protein (MAP) kinase pathways, which in turn, regulate the transcriptional factors *efg1* and *cph1*, respectively. These transcriptional factors, in turn, regulate the expression of important hyphal-specific genes such as *hwp1*, *als3* and *eap1* that are involved in filamentation [69–71]. In addition to hyphal morphogenesis, *efg1* is also reported to be crucial for biofilm formation, which was previously affirmed through the incompetence of *efg1* mutant to form robust biofilm [72]. Furthermore, *efg1* is also stated to be entailed in the production of secreted aspartic proteases in *C. albicans* [73]. In this study, 5HM2F considerably down regulated the crucial regulators of hyphal morphogenesis and virulence such as *ras1*, *efg1* and *cph1* in both *C. albicans* and mixed-species, which underscores the antibiofilm, antihyphal and antivirulence activity of 5HM2F witnessed in phenotypic assays.

Further, the hyphal-specific genes *hwp1*, *als3* and *eap1* are reported to mediate adhesion of *C. albicans* to abiotic surfaces & host epithelial cells, autoaggregation, co-aggregation with bacterial species, biofilm formation and invasion [70,74,75]. Notably, the positive role of *als3* in inter-kingdom interaction of *C. albicans* and *S. epidermidis* has been revealed by Beaussart et al. [63] through single-cell force spectroscopic study. Herein, 5HM2F turned down the expression level of these hyphal –specific genes, which concurs well with the impairments of 5HM2F treated cells observed in SEM analysis and *in vitro* bioassays such as cell adherence, auto-aggregation, co-aggregation and hyphal-bacterial

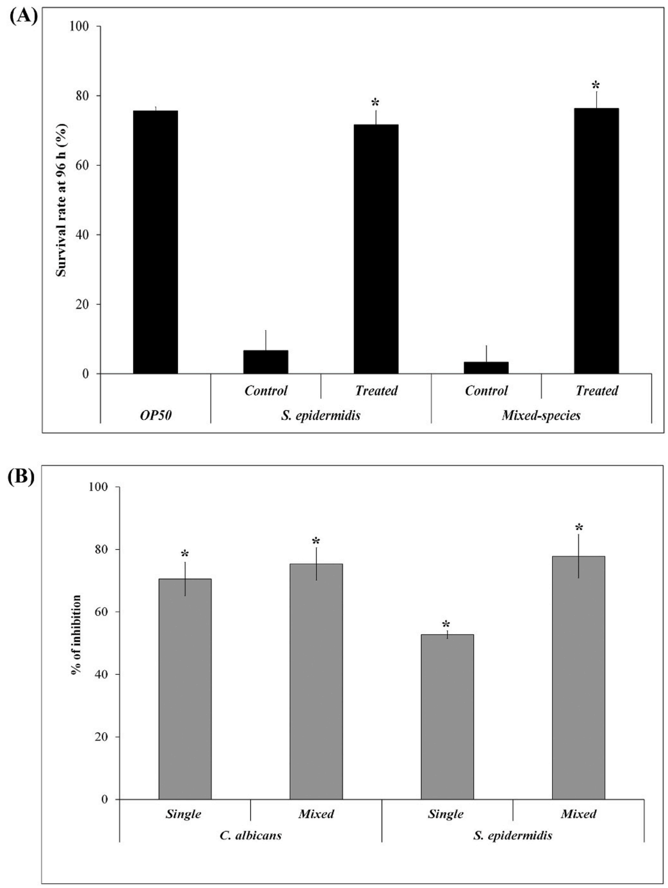

Fig. 9. *In vivo* assays using *C. elegans* model. (A) Bar graph depicting the ability of 5HM2F to enhance the survival rate of *S. epidermidis* and mixed-species infected worms. (B) Bar chart representing the impact of 5HM2F on microbial burden in mono-species and mixed-species infected worms. Error bars indicate standard deviation. * indicates statistical significance between control and 5HM2F treated samples ($p < 0.005$).

attachment assays. Furthermore, *ume6* is stated to be a crucial regulator, which mediates true hyphal extension in *C. albicans*. Importantly, *ume6* mutant was observed to be less virulent *in vivo* [76]. Thus, down regulation of filament-specific *ume6* upon 5HM2F treatment suggests the potential of 5HM2F to repress the virulence level of *C. albicans* and mixed-species. Alongside, 5HM2F was found to up regulate the expression of transcriptional repressor of filamentous growth *tup1* (Fig. 8A), which corroborates well with the antihyphal activity of 5HM2F observed against *C. albicans* and mixed-species. Altogether, phenotypic assays and qPCR analysis suggest that, 5HM2F attenuates biofilm formation and mixed-species interaction by interfering with yeast to hyphal transition and filamentation in *C. albicans* and mixed-species.

On the other hand, the expression level of *S. epidermidis* specific genes involved in initial adherence, aggregation, biofilm formation and accumulation were analyzed in single and mixed-state. Biofilm formation in *S. epidermidis* is reported to be mediated alternatively by *ica* dependent and *ica* independent mechanisms. In *ica* dependent mechanism, synthesis of PIA that aids intercellular adhesion and biofilm formation is expedited by the combined action of N-acetyl glycosamine transferases, *icaA* and *icaD*. In *ica* independent mechanism, *S. epidermidis* utilizes accumulation associated genes such as *aap* and *bhp* to build robust biofilm [77]. Down regulation of *icaA*, *icaD*, *aap* and *bhp* upon 5HM2F treatment defines the ability of 5HM2F to deter both *ica* dependent and *ica* independent biofilm formation of *S. epidermidis*. Amid this, adherence of *S. epidermidis* to host matrix proteins is mediated by genes encoding microbial surface components recognizing adhesive matrix molecules (MSCRAMM). In this study, 5HM2F was found to repress the expression of MSCRAMM associated genes such as *ebh*, *sdrG*, *sdrF* and *sdrH*, which specifies the ability of 5HM2F to perturb *S. epidermidis* interaction with host matrix proteins. Additionally, the bifunctional autolysin/adhesin *atlE* is stated to promote eDNA mediated biofilm formation and inter-kingdom interaction [18,78]. Importantly, *atl* (autolysin/adhesin) of *S. aureus* (a virulent cousin of *S. epidermidis*) has been reported to be one of the main adhesins to exert high affinity with *als3* expressing *C. albicans* in mixed-species biofilms [19]. Accordingly, down regulation of *atlE* expression and reduction of eDNA content in single and mixed-species biofilms upon 5HM2F treatment promulgates the inhibitory effect of 5HM2F against biofilm formation and *C. albicans* – *S. epidermidis* interaction. Besides, *S. epidermidis* virulence, invasive lifestyle and drug resistance are recounted to be facilitated by accessory gene regulator (*agr*) system, which regulates the secretion of virulent exoenzymes such as lipases and proteases [79]. Expression of *agrA* (response regulator of *agr* system) was found to be down regulated by 5HM2F (Fig. 8B), which is in line with the reduced exoenzymes production upon 5HM2F treatment in protease and lipase assays. Overall, qPCR study corroborates well with the *in vitro* bioassays and suggests that 5HM2F inhibits biofilm formation and mixed-species interaction by repressing the expression of *S. epidermidis* specific genes associated with initial adherence and cell aggregation.

In furtherance, the antibiofilm and antivirulence potential of 5HM2F was validated using *C. elegans*, which is a widely accepted simple eukaryotic model for studying drug toxicity and host-pathogen interactions [80,81]. Non-toxic nature of 5HM2F and its antivirulence potential against *C. albicans* was previously substantiated using *C. elegans* model [24]. In the present investigation, the antibiofilm and antivirulence potential of 5HM2F against *S. epidermidis* and mixed-species was studied *in vivo*. The ability of 5HM2F to enhance the survivability of worms infected with *S. epidermidis* and mixed-species (Fig. 9A) highlights the antivirulence potential of 5HM2F. Additionally, reduced *in vivo* colonization of *C. albicans*, *S. epidermidis* and mixed-species in the presence of 5HM2F, as confirmed by CFU assay (Fig. 9B) symbolizes the antibiofilm potential of 5HM2F. The plausible mode of action of 5HM2F speculated using *in vitro*, transcriptomic and *in vivo* studies is presented in Fig. 10.

5. Conclusion

In summary, the present investigation demonstrates the non-microbicidal antibiofilm and antivirulence efficacy of 5HM2F against *S. epidermidis* and mixed-species. *In vitro* bioassays and microscopic analyses revealed the competency of 5HM2F to mitigate yeast to hyphal transition, secreted hydrolases production, initial attachment, aggregation, and fungal-bacterial interaction. Further, 5HM2F mitigated the content of matrix components in mono-species and mixed-species biofilms, which is anticipated to be the contributing factor for enhanced susceptibility of 5HM2F treated cells to antibiotic and/antifungals. Consistent with phenotypic bioassays, qPCR analysis also explicated the ability of 5HM2F to turn down the expression level of genes associated with yeast to hyphal transition, initial attachment, aggregation, mixed-species interaction and virulence in both mono-species and mixed-species. 5HM2F recuperated the survivability of *S. epidermidis* and mixed-species infected *C. elegans*, which expounds the antivirulence potential of 5HM2F *in vivo*. Additionally, reduced microbial burden in 5HM2F treated worms substantiated the antibiofilm efficacy of 5HM2F. Altogether, the integrated *in vitro* and *in vivo* studies infer the efficacy of 5HM2F to reduce biofilm formation and inter-kingdom interactions of *C. albicans* and *S. epidermidis*. However, studies using higher eukaryotic models are indispensable before subjecting this antibiofilm 5HM2F to clinical investigations.

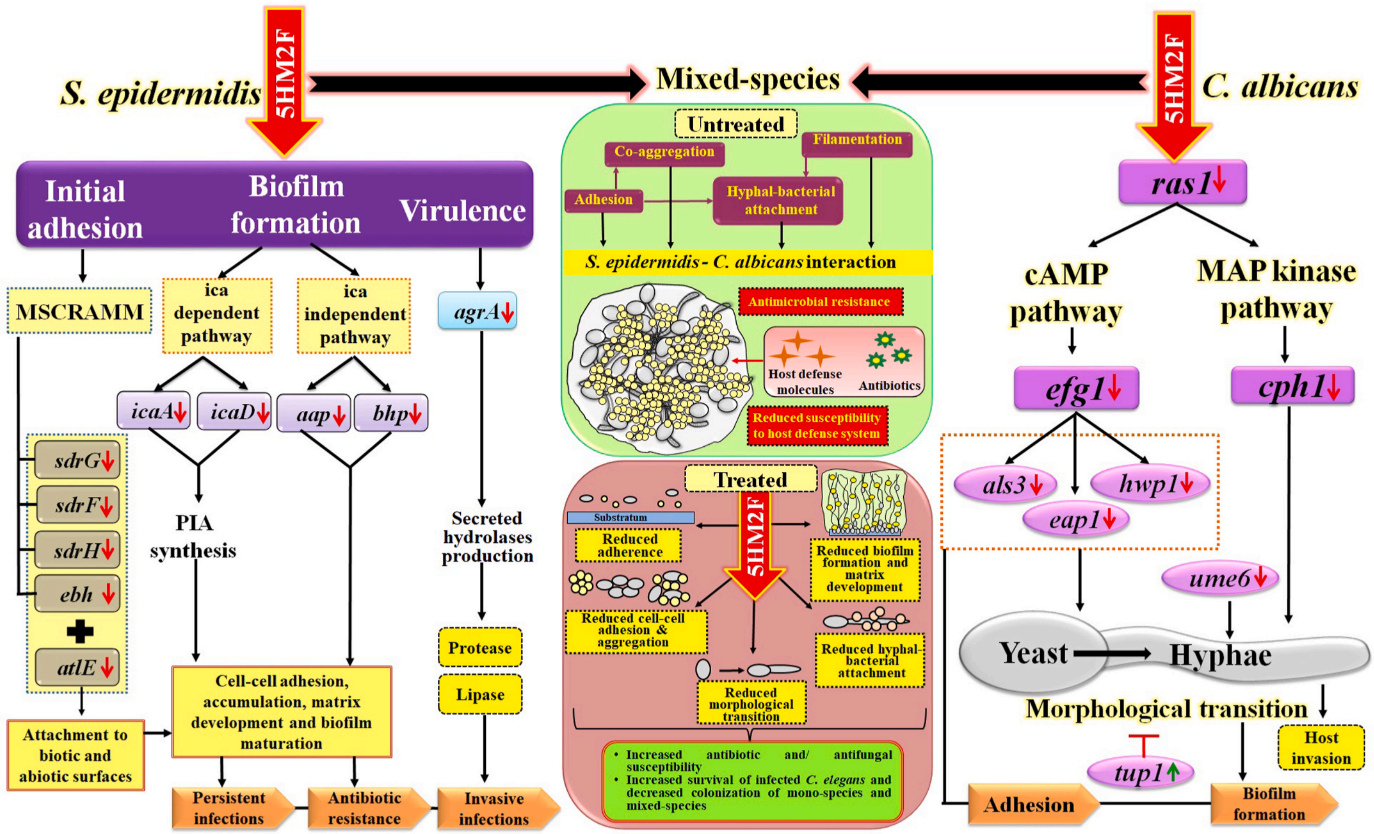

Fig. 10. Schematic illustration of plausible mode of antibiofilm action of 5HM2F against single and mixed-species of *S. epidermidis* and *C. albicans*. Red arrows indicate down regulation and green arrow represents up regulation of candidate genes. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Abbreviations

5HM2F - 5-hydroxymethyl-2-furaldehyde; AB - alamar blue; AGE - agarose gel electrophoresis; ANOVA - analysis of variance; BSA - bovine serum albumin; CFU - colony forming unit; CLSM - confocal laser scanning microscope; CV - crystal violet; eDNA - extracellular DNA; EPS - extracellular polymeric substances; FBS - fetal bovine serum; LM - light microscope; MBIC - minimum biofilm inhibitory concentration; MIC - minimum inhibitory concentration; MTP - microtitre plate; OD - optical density; PBS - phosphate buffered saline; SEM - scanning electron microscope.

Author contributions

TKS, GAS, TK & SKP- Conceptualization and formal analysis; TKS, TK & RS – Methodology; TKS – Data curation, validation, visualization, original draft (writing); GAS & SKP – Original draft (Review & Editing); SKP – Investigation and supervision. All authors have read and approved the final manuscript.

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